Case Study :-

Netflix Recommendation System

→ Collects user viewing data in real time.

→ kafka ingests streaming data

→ Spark processes large datasets.

→ ML model recommend shows based on behaviour.

Explanation of Each Step in the Big-lifecycle in view of the Case Study :-

1) Data Generation :-

Netflix users generate Continuous data through actions such as play, pause, rewind, Search etc.

2) Real-time data ingestion (Kafka) :-

Apache kafka ingests millions of user interaction events per second, ensuring Scalable data.

3) Data Storage :-

Incoming data is stored in distributed data lakes and warehouses in a Systematic manner.

4) Data Processing and transformation (spark) :-

Apache spark performs large-scale batch and streaming transformations like feature Extraction, aggregation etc.

5) Feature Engineering:-

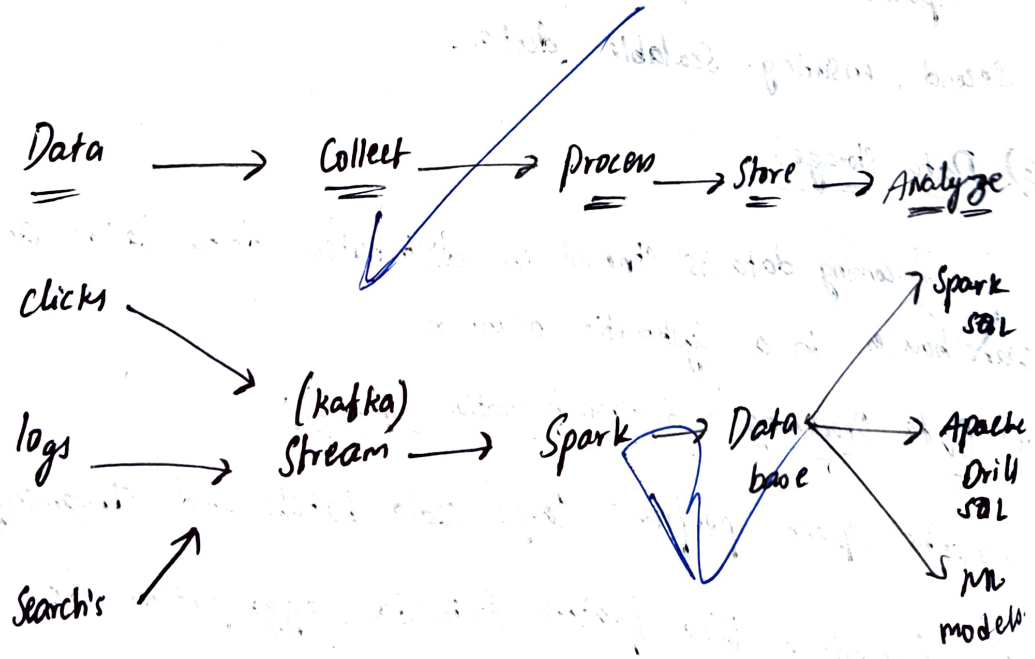
Procened data is converted into features such as watch history, genre preferences, time-of-day usage etc.

6) ML model training:-

ML models are trained on historical and behavioral data to predict user preferences.

7) Model Serving & Recommendations:-

Trained models serve low-latency predictions that personalize home pages, rankings and content suggestions for each user.

8) Feed back Loop:-

User interactions with recommendations are fed back into the system to continuously improve model accuracy.